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इंदिरा गांधी राष्ट्रीय मुक्त विश्वविद्यालय

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Course Code : BEVAE181

Course Title : Environmental Studies

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Tutor Marked Assignment
AECC on Environment Studies

Course Code: BEVAE-181
Assignment Code: BEVAE-181/TMA/2026
Maximum Marks: 100

Note: Attempt all questions. The marks for each question are indicated against it.

PART-A

1. "Sustainable development is an ideal goal towards which all human societies need to be moving" Justify the statement in about 250 words. (8)

2. Differentiate between the following terms by giving suitable examples in about 125 words each: (4x2=8)

(a) Genetic and species diversity
(b) Use and Non-use values of biodiversity

3. Answer the following questions in about 150 words each. (5x4=20)
   (a) The importance of Biomass has been increasing day by day in our surroundings among renewable resources. Explain it with suitable examples.
   (b) What is Disposal of waste? Why segregation of waste is needed?
   (c) Discuss any five consequences of deforestation with suitable examples.
   (d) Write a short note on Hydrological cycle with the help of a diagram.

4. Explain the human-environment relationship by taking examples of biotic and abiotic components? (7)

5. Critically evaluate non-polluting energy systems in India. Elucidate your answer with suitable examples in about 200 words. (7)

PART-B

6. Explain the following terms in about 60 words each: (2x4=8)
   (a) Greenhouse Effect
   (b) Climate Change
   (c) Agenda 21
   (d) Environmental Justice

7. Answer the following questions in about 150 words each. (5x4=20)
   (a) Explain the Importance of Grassland Ecosystem.
   (b) Define natural calamities and its types with suitable examples.
   (c) Describe issues emerges in enforcement of national environment legislations.
   (d) Explain the biocentrism and ecocentrism in context of human's attitude towards nature.

8. "Water Harvesting is one of the effective measures to combat drought." Explain this statement with suitable arguments. (7)

9. "Protected areas play a very important role in in-situ conservation of species." Elucidate the statement with respect to present day context in about 200 words. (7)

10. Differentiate between the primary and secondary pollutants. Explain how these pollutants are harmful for humans and environment in about 250 words. (8)

PART-A

Q.1 "Sustainable development is an ideal goal toward which all human societies need to be moving" Justify the statement in about 250 words.

Ans Introduction

Sustainable dellelopment refers to a pattern of development that meets the needs of the present without comprising the ability of future generations to meet their own needs It integrates economic growth, environmental protection and social equity This goal is essential because current development models have led to overexploitation of natural resources, climate change, biodiversity loss and environmental degradation. For example, exessive use of fossil fuels had caused global warming, while deforestation has disturbed ecological balance. Sustainable development promotes efficient resource use through renewable energy sources like solar & wind power, reducing dependence on finite resources It also emphasizes conservation practices such as afforestation, water harvesting and waste recycling.

Socially, it ensures inclusive growth by reducing poverty, improving healthcare, and ensuring equitable distribution of resources. Economically, it encourages long term rather than the short-term gains

Globally, initiatives like the sustainable Development Goals (SDGs) reflect this vision. countries are adopting green technologies, sustainable agriculture and eco-friendly industries.

Thus, Sustainable development is not just an ideal but a necessity. without it, environmental degradation will threaten human survival and economic stability. Therefore all societies must move toward sustainability to ensure a balanced and secure future.

Q2 Differentiate between the following terms by giving suitable examples in 125 words each.

(a) Genetic & species diversity.

Ans Genetic diversity refers to the variation of genes within a species. It enable species to adapt to environmental changes and resist diseases. For example, different varieties of rice or wheat show genetic diversity.

Species diversity, on the other hand refers to the variety of species within a particular region or ecosystem. It includes both the no. of species (richness) and their relative abundance (evenness). For example, a tropical rainforest has high species diversity compared to a desert.

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while genetic diversity operates at the micro leven (within species), species diversity operates at a broader ecological level. Both are crucial for ecosystem stability - genetic diversity ensures survival of species, while species diversity ensures ecosystem resilience

(b) User and Non-use values of Biodiversity

Ans Use values of biodiversity refer to the direct and indirect benefits human derive from biological resources. These include food, medicine, timber, fuel and ecosystem services like pollination and climate regulation. For example, plants used in medicine or forests providing timber.

Non-use values refer to the value people assign to biodiversity even without using it. These includes existence value (value of knowing species exist), aesthetic value, and ethical value. For example, conserving endangered species like tigers even if they are not directly used. Thus, use values are tangible and practical, while non-use values are intangible and ethical, but both are essential for conservation efforts.

Q.3 Answer the following questions in about 150 words
(a) The importance of Biomass has been increasing day by day in our surroundings among renewable resources. explain it with suitable examples.

Ans Biomass is organic material derived from plant and animals and is an important renewable energy source. Its importance has increased due to rising energy demands and environmental concerns.

Biomass can be used to produce biofuels such as biogas, biodiesel and ethanol. For example, cow dunk is used in rural areas to produce biogas for cooking. Agricultural waste like rice husk and sugarcane bagasse is used for electricity generation.

It helps reduce dependence on fossil fuels and lowers greenhouse gas emissions. Biomass energy is carbon-neutral as the carbon released during combustion is balanced by carbon absorbed during plant growth.

Additionaly, biomass promotes waste management by converting organic waste into energy, reducing landfill use.

Thus biomass is a sustainable, eco-friendly, economically viable energy source, especially for rural and developing regions.

(b) what is Disposal of waste? why segregation of waste is needed?

Sol: Waste disposal refers to the process of managing and eliminating waste minerals safely to minimise environmental and health hazards.
Segregation of waste is the process of separating waste into categories such as biodegradable, non-biodegradable, recyclable and hazardous waste. It is necessary because mixed waste is difficult to profess and leads to pollution.
For examples, biodegradable waste can be composted while plastic can be recycled. Hazardous wastes like batteries require special treatment.
Segregation improves recycling efficiency, reduces landfill burden, and prevents environmental contamination. It also helps in resource recovery and energy generation. Thus, proper waste segregation is essential for effective waste management and environmental sustainability.

(c) Discuss any five consequences of deforestation with suitable examples.

Ans: Deforestation leads to several environmental and ecological problems.

(1) Loss of Biodiversity: Many species lose thier habitat and become endangered or extinct (eg- Orangutans)

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2. climate change :- Tree absorbs CO_2, their removal increases greenhouse gases

3. soil erosion :- Roots bind soil, without them, soil is washed away.

4. Disruption of water cycle :- Reduced rainfall and increased drought conditions

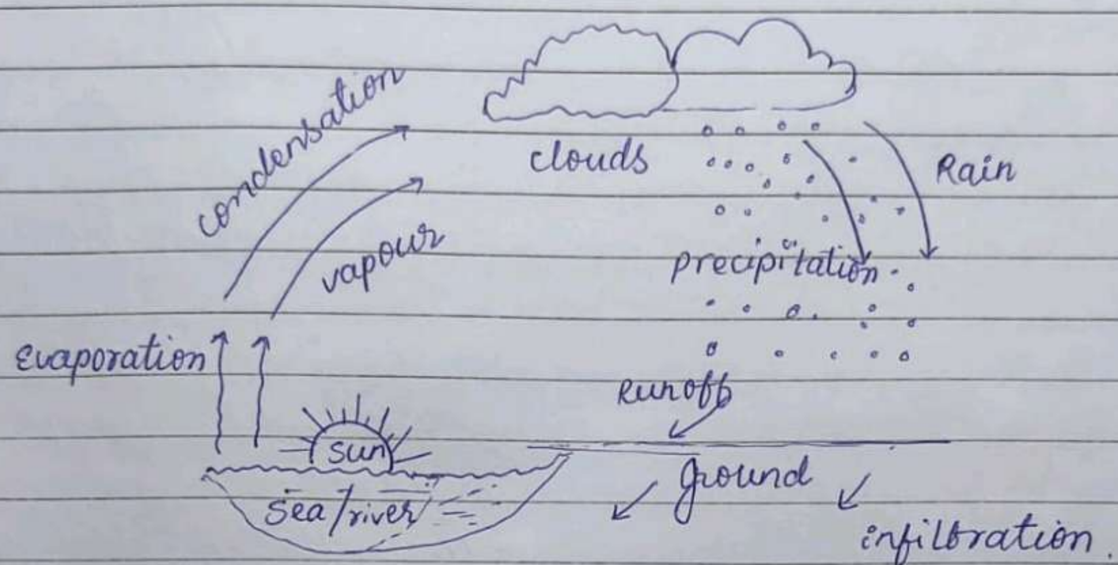
5. Flooding :- Lack of vegetation leads to increased surface runoff

For example, deforestation in the Amazon rainforest has significantly impacted global climate patterns. Thus, deforestation threatens ecological and human survival.

(d) write a short note on hydrological cycle?

Ans The hydrological cycle refers to the continuous movement of water between earth's surface and atmosphere.
It involves processes like
(a) Evaporation :- water from oceans and rivers turn into vapour.
(b) Condensation :- vapour form clouds.
(c) precipitation :- Rain, snow or hails falls to earth.
(d) Infiltration & Runoff :- water seeps into ground or flows into rivers.

This cycle maintains water balance and support life on earth.



Q.4. Explain the human-environment retalionship by taking examples of biotic and abiotic components.

Ans The human-environment relationship describes how humans interact with natural surroundings including both biotic and abiotic components. Biotic components includes plants, animals and microorganisms. Humans depends on them for food, oxygen, medicine and ecosystem services. For example, forest provide oxygend and regulate climate.

Abiotic components include air, water, soil, and minerals. Humans rely on these for survival and development. For eg. water is essential for drinking and agriculture.

However, human activities like industrialisation, deforestation and pollution have disturbed

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this relationship. overuse of natural resources leads to environmental degradation.
A balanced relationship is necessary, where humans use resources sustainably without harming ecosystem
Thus, human are both dependent on and responsible for protecting the environment.

Q.5 Critically evaluate non-polluting energy systems in India. Elucidate your answer with examples.

Ans Non-polluting energy system refer to clean and renewable energy sources that do not harm the environment. In India, these include solar, wind, hydro and biomass energy.
India has made significant progress in solar energy through initiatives like solarparks and rooftop solar systems. Wind energy is prominent in states like Tamil Nadu and Gujarat.
Hydropower provides a major share of renewable energy though it has ecological concerns like displacement and habitat loss. Biomass energy is widely used in rural areas.
Despite benefits like reduced carbon emissions and sustainability, challenges remain. These includes high initial costs, intermittency of solar and wind energy and infrastructure limitations.
Government initiatives like the National solar mission and promotion of electric vechiles

support clean energy growth.
critically, while non-polluting energy systems reduce environmental damage, their implementation must be balanced with social and ecological considerations.
Thus, non-polluting energy systems are essential for Indias sustainable future but require careful planning and investment.

PART-B

Q.6. Explain the following terms in about 150 words each.

(a) Greenhouse effect :- The greenhouse effect is the process by which gases like CO_2 and methane trap heat in earth's atmosphere, keeping it warm. while natural and necessary for life, excessive greenhouse gases due to human activities lead to global warming.

(b) climate change :- climate change refers to long terms changes in temperature and weather-patterns, mainly due to human activities like burning fossil fuels. It results in global warming, rising sea levels, and extreme weather events.

(c) Agenda 21 :- Agenda 21 is a global action plan adopted at the earth summit (1992) to promote sustainable development. It focuses on environmental protection, poverty reduction, and resource management.

(d) Enviromental justice :- It ensures fair treatment of all people regardless of race or income in environmental policies. It addresses issues where marginalized community face higher environmental risks.

Q.7. Answer the following question in about 150 words each.

ⓐ Explain the importance of grassland ecosystems.

Ans Grassland ecosystems cover about 40% of earth's land surface and play a crucial role in maintaining ecological balance.

- Biodiversity support → It supports unique flora and fauna. India grasslands host species like blackbucks, wolves, florican birds
- Carbon storage :- Grassland soils store significant amounts of carbon, helping regulate the global climate.
- Livestock and agriculture :- They provide grazing land for millions of livestock, supporting pastoral communities and dairy industries worldwide.
- water regulation :- Grassland roots prevents soil erosion, maintain water infiltration and help recharge groundwater.

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- prevention to desertification:- continous grass cover prevents soil from turning into desert especially in semi-Arid regions.

However, grasslands are amongst the most threatend ecosystems due to agricultural enroachment. overgrazing and urbanization. Their conservation is vital for ecological health.

(b) Define natural calamities and its types with suitable examples.

Ans A natural calamity is a severe, adverse event that results from natural process of the earth, causing significant damage to life, property and environment.

Types of natural calamities:-

(1) Geological calamities :- earthquakes, volcanic eruptions and tsunamis.

(2) Meteorological Calamities:- cyclones, hurricanes tornadoes, and thunderstorms.

(3) Hydrological calamities:- Floods and landslide caused by excessive rainfall.

(4) Climatological calamities:- Droughts, heat waves and wildfires.

(5) Biological calamities :- Epidemics, locust invasions, ~~exam~~.

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Natural calamities cannot be prevented, but their impacts can be minimised through early warning systems, disaster preparedness, and resilient infrastructure.

(c) Describe issue emerges in a enforcement of national environment legislations.

Ans India has a comprehensive framework of environmental laws including the Environment protection Act (1986), water Act (1974), Air Act (1981), wildlife protection Act (1972). However enforcement remains weak due to several issues.

(1) Institutional weakness.
(2) Political Interference
(3) Lack of public awareness
(4) Inadequate penalty
(5) slow judicial process
(6) corruptions.

Reforms such as e-monitoring systems, strengthening of NGT (National Green Tribunal) and increasing public participation are essential to improve enforcement.

(d) Explain the biocentrism and ecocentrism in context of humans attitude towards nature.

Ans Biocentrism and ecocentrism are two important ethical perspectives that define how

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humans should relate to nature.

Biocentrism:

Biocentrism holds that all living organisms have inherent value, regardless of their utility to humans. It places individual living beings at the center of ethical consideration. For example, a biocentric perspective would oppose killing even past insects for human convenience, as every life has worth.

Ecocentrism :-

Ecocentrism goes further by assisting that the entire ecosystem including non-living components like rivers, mountains, air has intrinsic value. It prioritizes the health of whole ecosystem over individual specis or human needs.

Q.8 "Water harvesting is one of the effective measures to combat drought" explain this statement with suitable arguments.

Ans Water harvesting refers to the collection and storage of rainwater and other forms of water for human use before it is lost through runoff or evaporation. In a drought prone country like India, It is one of the most practical and traditional solutions.

Arguments supporting the statement :-

(1) Augments water supply :- water harvesting structures like check dams, bunds, tanks and rooftop systems collect monsoon water and make it available dry spells.

(2) Recharges groundwater :- Harvested rainwater percolates into the ground, recharging aquifers and wells. This ensures water availability long after rains stop,

(3) Reduces Runoff and ~~Low cost~~ Soil Erosion :- By slowing down surface runoff, water harvesting structures prevent topsoil loss and allow more water to to infiltrate, improving soil moisture for crops.

(4) Decentralised and low cost :- Unlike large dams water harvesting can be done at the village or household level with minimal investment, making it accessible to poor farming communities

(5) Supports Agriculture :- stored water can be used for irrigation during dry months, enabling crop cultivation even during moderate drought conditions.

Q.9 "Protected areas play a very important role in in-situ conservation of species" Elucidate the statement with respect to present day context in about 200 words.

Ans Protected Areas and In-situ conservation:-

In-situ conservation refers to the protection of species in their natural habitats. protected areas such as National parks, wildlife sanctuaries, Biosphere Reserves, and Tiger reserves are the primary tools in-situ conservation.

Roles of Protected Areas:-

(1) Habitat Preservation.
(2) Species Recovery
(3) Ecological Processes
(4) Climate Refuge
(5) Research and Monitoring.

Present Day context:-

India has over 900 protected areas covering about 5% of its geographic area. The recent declaration of cheetah Reintroduction in Kuno National Park and expansion of marine protected areas shows growing commitment. However, challenges like human-wildlife conflicts, poaching, and climate change continue to test these systems. strengthening buffer zones, community involvement, and anti-poaching measures are critical for their continued effectiveness.

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Q.10 Differentiate between the primary and secondary pollutants. Explain how these pollutants are harmful for humans and environment in about 250 words.

Ans Primary pollutants:

primary pollutants are those that are directly emitted into the atmosphere from identifiable sources. They are harmful in their original form and directly impact air Quality.

Examples;→ carbon monoxide (CO), sulfur Dioxide (SO_2), particulate matter (PM 2.5, PM10), NO_x.

Secondary pollutants :-

secondary pollutants are not emitted directly. They form in the atmosphere through chemical reactions between primary pollutants and other atmospheric compounds in the presence of sunlight.

Ex - Ozone (O_3) forms when NO_x reacts with Volatile organic compounds (VOCs in sunlight).

Harmful Effects on Humans :-

(1) Co reduces the blood's oxygen-carrying capacity, causing headaches, dizziness and death at high concentrations.

(2) SO_2 and NO_x cause respiratory diseases like asthma, bronchitis, and lung cancer.

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(3.) PM 2.5 penetrates deep into lungs, causing cardiovascular and pulmonary diseases.

(4) Ground-level ozone irritates eyes, throat, and lungs.

Harmful Effects on Environment :-

(1) Acid rain damages forests, lakes and aquatic life. It also corrodes building and monuments.

(2) Smog reduces visibility, harm crops, damages ecosystem.

(3) Ozone depletion in stratosphere increases UV radiation reaching earth, harming plants and marine organisms.

(4) Particulates settle on leaves, blocking photosynthesis and reducing agricultural productivity.